Stata Task - INSPIRE Internship Program

INSPIRE Bocconi

June 2026

This is the Stata test for the INSPIRE Internship Program, the flagship initiative dedicated to enthusiastic and motivated Bocconi students who wish to gain first-hand experience in developing cutting-edge empirical research.

THE DATASET
This dataset comes from a school survey carried out in Tanzania. The respondent to this interview was the **head teacher** and the interview is split into two parts:

1. **School data** - This file contains the first part of the interview. Each row corresponds to a single interview with a head teacher, in which we collect some basic information about the school
2. **Teacher data** - This is the second part of the same interview. We ask the head teacher to **list** three key teachers of the school, which will eventually form the sample of individual interviews. We need to list only one respondent per each category:
   (a) One Head Teacher
   (b) One Deputy Head Teacher
   (c) One Senior Teacher

   Hence, this file contains 3 rows per school

INSTRUCTIONS - READ CAREFULLY!

- The folder that you have received contains all the materials required for this task
- **Before starting, take some time to familiarise with the dataset by consulting the codebook included in the materials provided. Also, read the introductory THE DATASET section, which contains important information**
- You will need to complete this task using Stata. Please submit a fully replicable .do file, alongside all the output that you are requested to save. Please include any text answer to specific questions as a comment in the do file.
- Please, put raw data, clean data and outputs in different folders, and all the folders in one zipped file. Make sure that the structure of the folder is clear and tidy.
- You will be graded not only on the answers you give, but also on the process and how you present your notes in the do-file at your answers (i.e., your code file and comments).
- Don't worry if you don't have time to finish all the tasks. It is more important to receive a fully annotated do-files, describing what you do step by step.
- You must send your output in a .zip folder to `INSPIRE@unibocconi.it`, with the object ***"INSPIRE Internship Program - Stata Test - Name Surname"***

You have 1.5 hours to complete this task - Good luck!

1 Stata Test - Group B

1. This survey is the Baseline of a Randomized Control Trial (RCT) where the treatment is a school-based intervention evaluating the impact of promoting the awareness of mental health in primary schools in Tanzania.
   What are the main features of an RCT? Why does a policymaker should consider setting-up a RCT, in this specific context? (1000 characters max) *Hint: answer to this in maximum 15 minutes, to allow you to complete the actual test*

2. Import the ***School data*** file. What basic logical checks would you carry out in this dataset? *Hint - Please use the attached codebook to familiarise with the variables and their meaning, including variable and value labels.*

3. Correct the data accordingly in Stata, as a result of your checks. Make sure to detail your changes with clear comments. *Hint - it might be worth looking at the comments left by interviewers*

4. Merge this dataset with the ***Admin Data*** file. This dataset contains the administrative data (names and characteristics) of each village. *Note - if possible, position the variables coming from the admin data file as first in the resulting merged dataset.*

5. The variable *type* has been imported as a string variable. Recode it as numeric variable and fully label this with the description and values specified in the codebook.

6. Create an ID variable for the region, called *regionid*. The code for Kilimanjaro is 1 and for Dodoma is 2

7. The variable *timebegun* has been imported as a string variable. Convert it in the date-time Stata format, and display it in an informative and appropriate way. *This variable is not required for completing the following steps, hence if you struggle on this command you should skip this*

8. What is the share of school with a scientific track in urban/rural/mixed settings? By "scientific track" we mean a school with either one of these:

   - Maths track
   - Science track
   - IT Technology track

   To calculate this share, create a new aggregate category **100 "SCIENTIFIC AGGREGATE"** - create the aggregate category, label it and tabulate.

9. Save the cleaned merged Admin+School dataset

10. Import the ***Teacher dataset***. What basic logical checks would you carry out in this dataset? Please use the attached codebook to familiarise with the variables and their meaning. Correct the data accordingly in Stata, as a result of your checks. Make sure to detail your changes with clear comments.

11. *Hint! Read again the instructions in the* ***The Dataset*** *section for this point.* Have the listing protocols been followed? Build a check for this and comment (i.e., in each school, we should have three respondents - one of each category as per listing protocols)

12. Merge this dataset with the cleaned Admin+School dataset you saved in point 7

13. In the educational system of Tanzania, different subjects are taught in different grades. In Grade 1 and Grade 2, the only subjects to be taught are Counting, Reading and Writing. No other subject is taught until the start of Grade 3. Build a check that looks into this. Please comment - are there inconsistencies in the data?

14. Generate a table summarising the mean, range, standard deviation, minimum, maximum, 5th/90th percentiles of **age and gender** of teachers, by type of school (private vs public). OPTIONAL BONUS - Export the table in either csv or latex table

15. What edits would you do to the dataset to make it more user friendly? Please refer to the codebook attached